

SOC 690S: Machine Learning in Causal Inference

Week 12 and 13: Heterogeneous Treatment Effect

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CATEs under Conditional Exogeneity

Motivation for Conditional Average Treatment Effects

- Traditional causal inference focuses on a single Average Treatment Effect (ATE).
- Yet, individuals and subgroups may respond differently to treatment
- Estimating heterogeneous effects, or Conditional Average Treatment Effects (CATEs), enable researchers and policymakers to tailor interventions and understand for whom a policy works best

Setup for Conditional Average Treatment Effects

- We assume CIA under a full set of observed, potentially high-dimensional covariates Z_i

$$D_i \perp (Y_i(1), Y_i(0)) \mid Z_i$$

- Conditional on covariates Z_i , treatment is as good as randomly assigned
- We assume low-dimensional covariates X_i summarizes heterogeneity of the ATE conditioning on Z_i

$$\tau_0(X_i) = E [E[Y_i \mid D_i = 1, Z_i] - E[Y_i \mid D_i = 0, Z_i] \mid X_i]$$

where X_i is a subset of covariates Z_i

Parametric Estimation of CATEs

Parametric Estimation of CATEs

- The *parametric* estimation of $\tau_0(X)$ is straightforward
- The basic idea of *parametric* estimation includes two steps
 - ① A *unit*-specific estimation of ATE

$$Y_i(\eta_0) = E[Y_i \mid D_i = 1, Z_i] - E[Y_i \mid D_i = 0, Z_i]$$

where η represents the true nuisance CEF of Y given D and Z

- ② A function of $Y_i(\eta)$ against the same *unit*-specific covariates X

$$\tau_0(X_i) = E[Y_i(\eta_0) \mid X_i]$$

Parametric Estimation of CATEs: First Step

- To estimate unit-specific ATE $Y_i(\eta_0)$, we borrow the idea from *Interactive Regression Model* (IRM), which allows non-linear treatment effect heterogeneity across the high-dimensional Z_i
- By design of IRM (and AIPW), the estimand is doubly robust and Neyman orthogonal
- More specifically, we assume the underlying DGP to be fully non-linear, with true nuisance function g_0 and m_0

$$Y_i = g_0(D_i, Z_i) + \epsilon_i$$

$$D_i = m_0(Z_i) + \mu_i$$

Parametric Estimation of CATEs: First Step

- Under IRM, the estimation of $Y_i(\eta_0)$ is essentially the same as DiD, but using Y_i instead of ΔY_i , and the target estimand is ATE
- Specifically, we target a *doubly robust* ATE estimand—robust signal—in the form of *robust signal*

$$Y_i(\eta_0) = H_i(m_0)(Y_i - g_0(D_i, Z_i)) + g_0(1, Z_i) - g_0(0, Z_i)$$
$$H_i(m_0) = \frac{D_i}{m_0(Z_i)} - \frac{1 - D_i}{1 - m_0(Z_i)}$$

Parametric Estimation of CATEs: Generic DML for First Step

- Similar to DML DiD, there are two nuisance functions in *robust signal*— $m_0(\cdot)$ and $g_0(\cdot)$ —that may be estimated by DML
- The generic DML procedure applies to the estimation of the *robust signal*

Parametric Estimation of CATEs: Generic DML for First Step

- Similar to DML DiD, there are two nuisance functions in *robust signal*— $m_0(\cdot)$ and $g_0(\cdot)$ —that may be estimated by DML
- The generic DML procedure applies to the estimation of the *robust signal*
 - ① Partition data indices into k folds of approximately equal size
 - ② For each fold $k = 1, \dots, K$, compute ML estimators $\hat{g}_{[k]}(D_i, Z_i)$ and $\hat{m}_{[k]}(Z_i)$ of $g_0(\cdot)$ and $m_0(\cdot)$ in the left-out k -th fold of the data. For $i \in I_k$

$$Y_i(\hat{\eta}_{[k]}) = H_i(Y_i - \hat{g}_{[k]}(D_i, Z_i)) + \hat{g}_{[k]}(1, Z_i) - \hat{g}_{[k]}(0, Z_i)$$

$$H_i = \frac{D_i}{\hat{m}_{[k]}(Z_i)} - \frac{1 - D_i}{1 - \hat{m}_{[k]}(Z_i)}$$

Parametric Estimation of CATEs: Second Step

- We use the DML-estimated *robust signal* $Y_i(\hat{\eta})$ and a Best Linear Approximation (BLA) to estimate CATEs
- Specifically, the CATE function $\tau_0(X_i)$ may be parametrically approximated by a linear function

$$\tau_0(X_i) = E[Y_i(\eta_0) \mid X_i] \approx p(X_i)' \beta_0$$

where $p(X_i)$ is a low-dimensional dictionary (*e.g.*, group dummies, polynomials, or splines)

- β_0 is chosen to minimize the loss function

$$\beta_0 = \arg \min_{\beta} E[(\tau_0(X_i) - p(X_i)' \beta)^2] = \arg \min_{\beta} E[(Y_i(\eta_0) - p(X_i)' \beta)^2]$$

Parametric Estimation of CATEs: Second Step

- We may use $p(X_i)$ to represent groups, such as, income quintiles, polynomials, or splines (smooth dependence on X_i)
- For example, we may use

$$p(X_i) = (\mathbb{1}\{X_i \in R_1\}, \dots, \mathbb{1}\{X_i \in R_K\})'$$

- And $\hat{\beta}$ recovers the *group average treatment effects* (GATE)
- These approximations yield interpretable summaries of heterogeneity

$$\hat{\beta} = \left(\frac{1}{n} \sum_{i=1}^n p(X_i)p(X_i)' \right)^{-1} \frac{1}{n} \sum_{i=1}^n p(X_i)Y_i(\hat{\eta})$$

Parametric Estimation of CATEs: Generic DML

- Under regularity conditions (Semenova and Chernozhukov 2021)

$$\sqrt{n} \left(\hat{\beta} - \beta_0 \right) \overset{a}{\sim} \mathcal{N} \left(0, \hat{\Omega} \right)$$

with

$$\hat{\Omega} = \hat{Q}^{-1} \left[\mathbb{E}_n p(X_i) p(X_i)' (Y_i(\hat{\eta}) - p(X_i)' \hat{\beta})^2 \right] \hat{Q}^{-1}$$

and $\hat{Q} = \mathbb{E}_n [p(X_i) p(X_i)']$

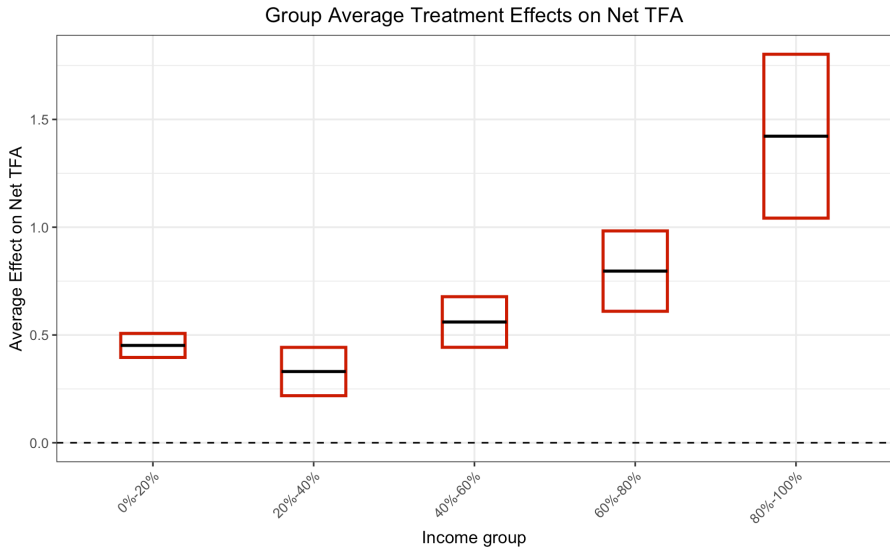
Parametric Estimation of CATEs: Case of 401(k)

- We consider again the effect of 401(k) eligibility on net total financial assets controlling for household characteristics
- We consider heterogeneity of the effect as a function of income
- We consider GATEs based on coarse income categories (see the coding file for CATEs based on polynomial terms in income)

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- We use 2-fold cross-fitting and default Random Forest in estimating the two nuisance functions

Parametric Estimation of CATEs: Case of 401(k)



Nonparametric Inference on CATEs with Causal Forests

Motivations for Nonparametric Inference

- In *parametric* approaches, researchers pre-specify the set of covariates X_i on which to condition on with a certain functional form in X_i when estimating the CATE
- In practice, the true sources of treatment-effect heterogeneity within X_i may be complex or nonlinear
- The *generalized random forest* (GRF) keeps the choice of X_i fixed but learns, in an *algorithmically honest* way, how to partition the space of X_i to maximize heterogeneity in the estimated treatment effects

Generalized Random Forests: Setup

- A *nonparametric* approach to estimating the CATE is to express it as solving a local conditional moment restriction

$$E[m(W_i; \tau_0(X_i), \eta_0) \mid X_i] = 0$$

- W_i collects all observed variables (Y_i, D_i, Z_i) , where X_i (the effect-modifying features) can be a subset of Z_i or equal to Z_i
- η_0 denotes nuisance functions
- $\tau_0(X_i)$ is the CATE given X_i
- The *generalized random forest (GRF)* estimates $\tau_0(X_i)$ nonparametrically by learning local neighborhoods in X_i that satisfy this conditional moment, while maximizing the induced heterogeneity in $\tau(X_i)$ across the partitioned space of X_i in each bootstrapped subsample from Random Forest

Generalized Random Forests: Estimation

- A straightforward practice called *Doubly Robust Forest* that links our previous *robust signal* is to find the best partitioning rule through Random Forest on the vector space X_i for leaf- or node-specific mean *robust signal* to form $\tau(X_i)$
- The induced heterogeneity in $\tau(X_i)$ across the partitioned space of X_i is maximized
- Formally, after *global* estimation of the nuisance functions $g(\cdot)$ and $m(\cdot)$, within node P ($X_i \in P$)

$$m(W_i; \tau_P, \eta) = \frac{D_i - m(Z_i)}{m(Z_i)(1 - m(Z_i))} (Y_i - g(D_i, Z_i)) + g(1, Z_i) - g(0, Z_i) - \tau_P(X_i)$$

- The solution $\tau(X_i)$ is essentially the same as constructing the *robust signal* in a cross-fitting approach on the *whole sample*, and calculate the mean *robust signal* within P

Generalized Random Forests: Estimation

- Let C_1 and C_2 denote the child nodes that will be created from some candidate split, with sample size n_1 and n_2
- A proxy criterion that targets maximizing between-node heterogeneity is to maximize $n_1\tau_{C_1}^2 + n_2\tau_{C_2}^2$
 - In practice, the number and depth of nodes (leaves) are not fixed in advance; they're determined adaptively during tree construction
- For this specific tree b with two child nodes, the CATE at $X_i = x$ is estimated as $\hat{\tau}_{L_b(x)}$, where $L_b(x)$ represents the leaf node x falls in (C_1 or C_2)
- For B such trees, the CATE at $X_i = x$ is

$$\text{CATE}(x) = \frac{1}{B} \sum_{b=1}^B \hat{\tau}_{L_b(x)}$$

Generalized Random Forests: Estimation

- More formally, once we have these partitions over B trees in the forest, let $L_b(x)$ denote the leaf node that a sample with covariates x falls in at tree b and $|L_b(x)|$ the number of units in that leaf
- The CATE at $X_i = x$ is defined as solving the (weighted) empirical average version of the moment condition

$$\begin{aligned} 0 &= \frac{1}{B} \sum_{b=1}^B \left[\frac{1}{|L_b(x)|} \sum_{i \in L_b(x)} m(W_i; \hat{\tau}_{L_b(x)}, \hat{\eta}) \right] \\ &= \frac{1}{B} \sum_{b=1}^B \left[\sum_{i=1}^n \frac{\mathbb{1}\{L_b(x) = L_b(X_i)\}}{|L_b(x)|} m(W_i; \hat{\tau}_{L_b(x)}, \hat{\eta}) \right] \end{aligned}$$

- This is equation 14.3.3 in CML

Generalized Random Forests: Estimation Ingredients

- The CATEs estimated through GRF are asymptotically normal with valid confidence interval if the following criteria are satisfied
 - *Honesty*: a separate sample is used to construct the structure of the tree and a separate sample is used to calculate the estimates at the leaf nodes of the tree
 - The reason for having *honesty* in estimation is similar to cross-fitting
 - We do not want to use the same set of observations for model selection and estimation to address “adaptive bias”
 - For example, the split rule will *chase noise*—they pick splits that maximize apparent heterogeneity even if its random
 - The estimated leaf-level CATE will then be likely very biased, as the same random noise that drives the split affects the local estimate

Generalized Random Forests: Estimation Ingredients

- *Balancedness*: every split should leave at least a $\rho \geq 0.2$ fraction of the samples on each side
- Note that this ρ speaks to the parent node's fraction in each split, not the total sample
- *Random Feature Split*: Every feature in X_i should have a probability of at least π/d to be chosen on each split, where d is the number of features in X_i
- *Fully Grown*: the tree should be grown fully, such that the number of samples that fall in every leaf should be at most some small constant

Generalized Random Forests: Estimation Ingredients

- *Sub-Sampling*: unlike typical Random Forest methods that use bootstrap samples of the same size as the original samples and drawn with replacement, the adapted forests in GRF use sub-samples *without* replacement and of a smaller size $s \ll n$ than the original sample on each tree
- The size of the sub-sample needs to be chosen carefully for the validity of the confidence intervals and should be of the order of $n^{\frac{\alpha d}{\alpha d + 1}}$, where
$$\alpha = \frac{\log(1/\rho)}{\pi \log(1/(1-\rho))}$$
- In practice, packages like grf default to $s \approx n/2$ for each bootstrapped (without replacement) tree
- Within such trees, *honest* estimation is used (meaning separate sample estimation within sub-sampled tree)

Causal Forest as a GRF for CATE

- An empirically popular variant of causal forests that uses the GRF formulation is the Causal Forest
- For asymptotics and for simplicity (Wager and Athey 2018; Athey et al. 2019), the approach is valid only if $Z = X$ and is low-dimensional
- Instead of estimating a global *robust signal* that is already Neyman orthogonal, Causal Forests estimate CATE within each leaf after partition in a DML approach in Partial Linear Model (PLM) for orthogonality
- Partition rules are made similarly by maximizing between-leaf variance in CATE

Causal Forests as a GRF for CATE

- Specifically, let $\hat{Y}$ and $\hat{D}$ represent globally residualized variables that can be estimated through ML with cross-fitting

$$\hat{Y} = Y - E[Y | Z]; \quad \hat{D} = D - E[D | Z]$$

- Within each leaf $L_b(x)$, the CATE $\tau_{L_b(x)}$ is defined by solving the moment condition

$$E[m(W; \tau(x), \hat{\eta}) | X = x] = E[(\hat{Y} - \tau(x)\hat{D})\hat{D} | X = x] = 0$$

- The same aggregation method as in GRF applies
- For B such trees, the CATE at $X_i = x$ is

$$\text{CATE}(x) = \frac{1}{B} \sum_{b=1}^B \hat{\tau}_{L_b(x)}$$

Causal Forests as a GRF for CATE

- Note that X can still have multiple dimensions, so that to visually show CATE as a function of one particular variable X_1
- A common approach is through *partial dependence* plot, where we hold other covariates $X_2, \dots, X_k$ at their mean or median values
- For each tree b , there exists a leaf where X_1 takes the *varying* value in plot, while other covariates take the *fixed* values $\bar{X}_2, \dots, \bar{X}_k$
- Because of decorrelation, some leaves may not contain X_1 or some of the covariates
- We use leaves that contain X_1 and part of the covariates at their mean or median value and define CATE at $x = (x_1, (\bar{X}_2, \dots, \bar{X}_k))$ as

$$\frac{1}{\sum \mathbb{1}\{x_1 \in x\}} \sum_{b=1}^B \hat{\tau}_{L_b(x)} \mathbb{1}\{x_1 \in x\}$$

Causal Forest: When Treatment is Continuous

- While the DR-AIPW approach in HTE does not allow the treatment to be continuous, there is now solid econometric and ML theory for *continuous-treatment* Causal Forest estimation of heterogeneous dose-response functions
- We again assume an IRM-like DGP, where

$$Y = \mu_0(X, D) + \epsilon$$
$$\tau(X) = \left. \frac{\partial E[Y \mid X, D]}{\partial D} \right|_{D=d}$$

- The moment condition used in GRF has the same form of

$$E[m(W; \tau(X), \eta_0) \mid X] = 0$$

with the standard orthogonal score defined as (Chernozhukov et al. 2021; Athey et al. 2019)

$$m(W; \tau, \eta) = (D - e(X))(Y - g(X) - \tau(D - e(X)))$$